

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

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⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

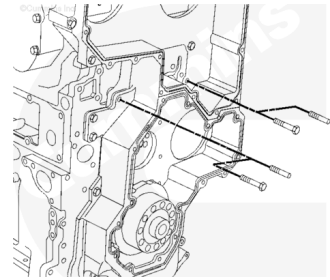
- Before removing gears, read Engine Base Timing. Refer to Procedure 001-088 in Section 1 (</qs3/pubsys2/xml/en/procedures/10/10-001-088-tr.html>).
- Remove the water pump drive belt. Refer to Procedure 008-003 in Section 8 (</qs3/pubsys2/xml/en/procedures/10/10-008-003.html>).
- Remove the alternator drive belt. Refer to Procedure 013-005 in Section 1 (</qs3/pubsys2/xml/en/procedures/10/10-013-005.html>).
- Remove the fan hub assembly. Refer to Procedure 008-036 in Section 8 (</qs3/pubsys2/xml/en/procedures/10/10-008-036-tr.html>).
- Remove the crankcase breather tube. Refer to Procedure 003-018 in Section 3 (</qs3/pubsys2/xml/en/procedures/10/10-003-018.html>).
- Remove the alternator. Refer to Procedure 013-001 in Section 13 (</qs3/pubsys2/xml/en/procedures/10/10-013-001-tr.html>).
- Drain coolant. Refer to Procedure 008-018 in Section 1 (</qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html>).

- Remove the crankcase breather tube. Refer to Procedure 003-001 in Section 3
(/qs3/pubsys2/xml/en/procedures/10/10-003-001-tr.html).
- Remove the upper and lower gear covers. Refer to Procedure 001-079 in Section 1
(/qs3/pubsys2/xml/en/procedures/10/10-001-079-tr.html) and Refer to Procedure 001-080 in Section 1
(/qs3/pubsys2/xml/en/procedures/10/10-001-080-tr.html).
- Remove the water pump. Refer to Procedure 008-062 in Section 8 (/qs3/pubsys2/xml/en/procedures/10/10-008-062-tr.html).
- Remove the valve and injector camshaft scissor gear. Refer to Procedure 001-088 in Section 1
(/qs3/pubsys2/xml/en/procedures/10/10-001-088-tr.html).
- Remove the adjustable idler gear. Refer to Procedure 001-088 in Section 1
(/qs3/pubsys2/xml/en/procedures/10/10-001-088-tr.html).
- Remove the air compressor. Refer to Procedure 012-014 in Section 12 (/qs3/pubsys2/xml/en/procedures/10/10-012-014-tr.html).
- Remove the lower idler scissor gear. Refer to Procedure 001-088 in Section 1
(/qs3/pubsys2/xml/en/procedures/10/10-001-088-tr.html).
- Remove the crankshaft position sensor and capscrews from front of oil pan.
- Remove the retaining brace bracket. Refer to Procedure 001-081 in Section 1
(/qs3/pubsys2/xml/en/procedures/10/10-001-081.html).
- Remove the accessory bracket. Refer to Procedure 001-082 in Section 1
(/qs3/pubsys2/xml/en/procedures/10/10-001-082.html).

Remove



The component weighs 23 kg [50 lb] or more. To reduce the possibility of injury, use a hoist or get assistance to lift the component.



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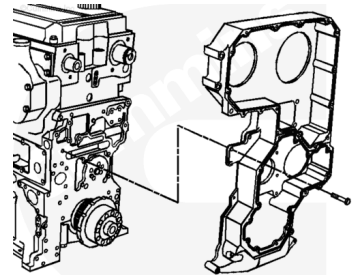
Remove two capscrews and install two guides.

Remove the remaining 18 capscrews.

Remove the housing and gasket.

Discard the gasket.

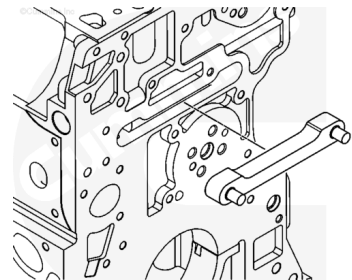
Remove the guide studs.



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⚠ CAUTION ⚠

Immediately upon removal of the front gear housing a plug must be inserted into the cylinder block oil passage drilling. Failure to insert the oil passage plug can result in a bearing failure, crankshaft failure, or both.



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The reusable plug, Part Number 4918320, is included in protective plug kit, Part Number 4918319. This plug is necessary to prevent debris from entering the lubrication system during the repair.

Insert the plug into the front oil passage.

Firmly push the plug into the oil passage to prevent all debris from entering the lubrication system when cleaning the gasket surface.

Note : Blocks built after 2006 will require the newer reusable plug, Part Number 4918874.

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

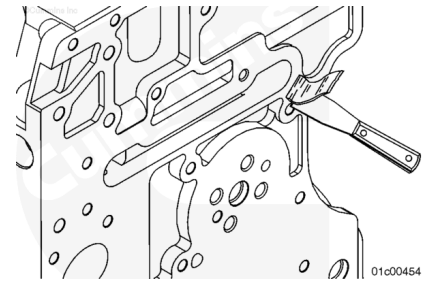
When using a steam cleaner, wear protective clothing and safety glasses, or a face shield. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

If the oil passages are not plugged before cleaning, a crankshaft failure can occur.

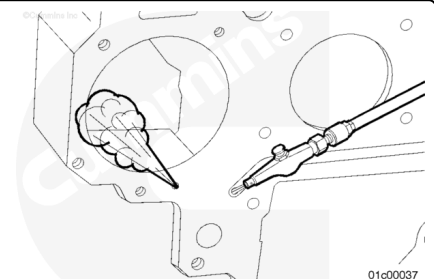


Clean gasket surfaces by hand with a gasket scraper or cleaning pad, Part Number 3823258. If the gasket material residue can **not** be felt with a finger, the surface is ready to accept the new gasket.

Use steam or solvent to clean the gear housing and dry with compressed air.

⚠ WARNING ⚠

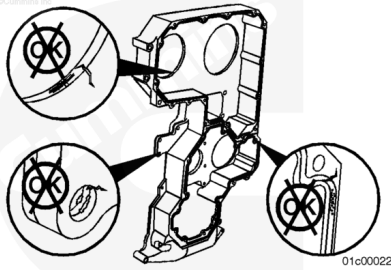
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Use compressed air to clear the oil galley to the air compressor and oil pump of any obstructions.

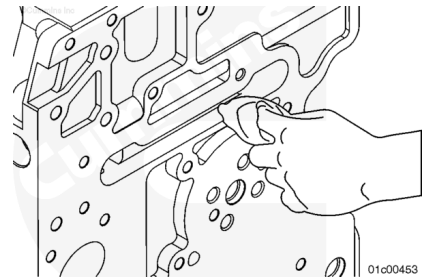
Inspect the gear housing for cracks or other damage.

Inspect the seal grooves of the housing for debris or damage.



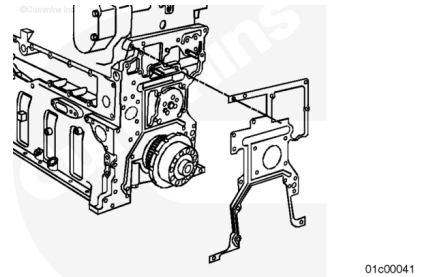
Install

Remove the protective plug from the cylinder block oil passage. Make sure no debris enters the lubrication system. Use a clean shop towel to carefully wipe out the oil passage and inspect to make sure no debris is left in the passages.

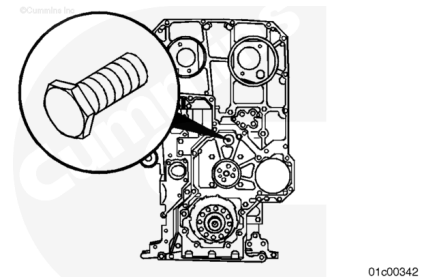


Install two guide studs.

Install a new gear housing mounting gasket.



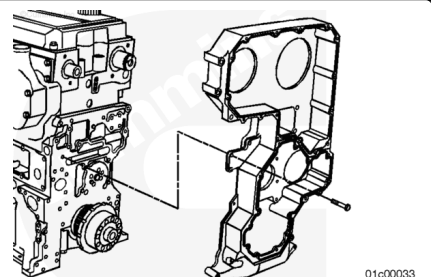
The gear housing has the words "Special Capscrew" cast into the housing at the number 1 mounting location (see illustration). This capscrew **must** be the short head capscrew to prevent contact between the capscrew and the idler gear.



⚠ WARNING ⚠

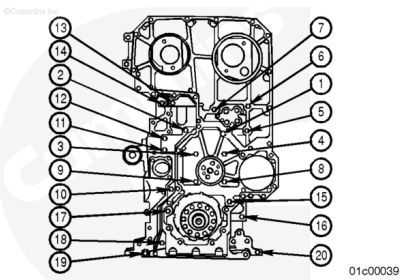
This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.

Install the gear housing.



Install the 18 mounting capscrews.

Four different length capscrews are utilized on the front gear housing. Use the table below to identify the capscrews and their mounting locations.



Capscrew Mounting Location		
Location	Size	Quantity
14	M10 x 1.5 x 93	1
18, 19, 20	M10 x 1.5 x 70	3
1	M10 x 1.5 x 21.8 (Special Short Head)	1
All remaining	M10 x 1.5 x 2.5	15

Remove the guide studs and install two capscrews.

Tighten the capscrews in the sequence shown.

Torque Value:

All Capscrews Except (1)

- 1. 20 n•m [15 ft-lb]
- 2. 72 n•m [53 ft-lb]

Torque Value:

Capscrew (1)

- 1. 20 n•m [15 ft-lb]
- 2. 56 n•m [41 ft-lb]

Finishing Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



- Install the accessory bracket. Refer to Procedure 001-082 in Section 1. (/qs3/pubsys2/xml/en/procedures/10/10-001-082.html)

- Install the retaining brace bracket. Refer to Procedure 001-081 in Section 1.
(/qs3/pubsys2/xml/en/procedures/10/10-001-081.html)
- Install the crankshaft position sensor. Use the following procedure in the Signature, ISX, and QSX15 Service Manual, Bulletin 4021334. Refer to Procedure 019-042 in Section 19. (/qs3/pubsys2/xml/en/procedures/70/70-019-042.html)Install the capscrews at the front of the oil pan.
- Install the lower idler scissor gear. Refer to Procedure 001-088 in Section 1.
(/qs3/pubsys2/xml/en/procedures/10/10-001-088-tr.html)
- Install the air compressor. Refer to Procedure 012-014 in Section 12. (/qs3/pubsys2/xml/en/procedures/10/10-012-014-tr.html)
- Install the adjustable idler gear. Refer to Procedure 001-088 in Section 1.
(/qs3/pubsys2/xml/en/procedures/10/10-001-088-tr.html)
- Install the valve and injector camshaft scissor gear. Refer to Procedure 001-088 in Section 1.
(/qs3/pubsys2/xml/en/procedures/10/10-001-088-tr.html)
- Install the water pump. Refer to Procedure 008-062 in Section 8 (/qs3/pubsys2/xml/en/procedures/10/10-008-062-tr.html).
- Install the upper and lower gear covers. Refer to Procedure 001-079
(/qs3/pubsys2/xml/en/procedures/10/10-001-079-tr.html) and Refer to Procedure 001-080 in Section 1.
(/qs3/pubsys2/xml/en/procedures/10/10-001-080-tr.html)
- Install the crankcase breather tube. Refer to Procedure 003-001 in Section 3.
(/qs3/pubsys2/xml/en/procedures/10/10-003-001-tr.html)
- Fill coolant. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html)
- Install the alternator. Refer to Procedure 013-001 in Section 13. (/qs3/pubsys2/xml/en/procedures/10/10-013-001-tr.html)
- Install the crankcase breather tube. Refer to Procedure 003-018 in Section 3.

(/qs3/pubsys2/xml/en/procedures/10/10-003-018.html)

- Install the fan hub assembly. Refer to Procedure 008-036 in Section 8. (/qs3/pubsys2/xml/en/procedures/10/10-008-036-tr.html)
- Install the alternator drive belt. Refer to Procedure 013-005 in Section 13.
(/qs3/pubsys2/xml/en/procedures/10/10-013-005.html)
- Install the water pump drive belt. Refer to Procedure 008-003 in Section 8.
(/qs3/pubsys2/xml/en/procedures/10/10-008-003.html)
- Operate the engine to normal operating temperature and check for leaks.